

National Digital Platform for Research, Policy Innovation, and Evidence-Based Land Governance

Background:Land is a finite and strategic resource that underpins economic development, environmental sustainability, food security, urban expansion, and social equity. Effective land governance is therefore critical to achieving sustainable development goals and supporting India's rapidly evolving socio-economic landscape. However, the land administration ecosystem in India remains largely implementation-oriented, with limited institutional focus on applied research, policy experimentation, and evidence-based innovation.

Description:Develop a comprehensive National Digital Platform for Research and Policy Innovation that promotes applied research, policy experimentation, knowledge sharing, and evidence-based decision-making in land governance.

The platform should function as a centralized repository and collaborative ecosystem that integrates datasets, research publications, policy documents, geospatial information, analytical tools, and case studies from various government departments, academic institutions, and research organizations.

Scope of the Study: There is a pressing need for a dedicated digital platform that serves as a national knowledge ecosystem for researchers, policymakers, government agencies, academic institutions, and industry experts to collaborate, conduct research, evaluate policies, and develop innovative solutions for strengthening land governance across the country.

Aspect	Scope
Existing Research Ecosystem	Study the current landscape of land administration research, policy development, and institutional knowledge management in India.
Stakeholders	Ministry of Rural Development, State Governments, Academic Institutions, Research Organizations, Think Tanks, Policy Makers, Survey Agencies, GIS Experts, NGOs, and International Organizations.
Data Analytics	Utilize AI, Machine Learning, and statistical tools to analyze land administration data and generate evidence-based insights for policy formulation.
Geospatial Integration	Integrate GIS, remote sensing, satellite imagery, and spatial analytics for land use planning, climate resilience, and infrastructure development.
Dashboard Reporting	& Provide interactive dashboards, visualization tools, policy indicators, research metrics, and customized reports for administrators and researchers.
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Innovation Challenges	& Support hackathons, innovation challenges, pilot projects, and research initiatives to encourage innovative solutions for land governance.

Problems: Emerging challenges such as climate change, rapid urbanization, urban-rural land transitions, increasing land disputes, sustainable land use planning, geospatial governance, and digital transformation require continuous research and innovative policy interventions. Despite

the availability of vast datasets generated through land records, cadastral surveys, satellite imagery, GIS platforms, and government programmes, these resources remain underutilized for generating actionable insights and supporting informed policymaking.

Expected Solution: The proposed solution should be a secure, scalable, AI-enabled National Research and Policy Innovation Platform that strengthens evidence-based land governance through collaborative research, advanced analytics, and digital knowledge management.

The platform should provide:

7. A centralized digital repository for land governance research, policy papers, datasets, legal documents, and case studies.
8. AI-powered search and recommendation engine for discovering relevant research and policy resources.
9. Collaborative workspaces for researchers, policymakers, academic institutions, and government agencies.
10. Interactive GIS-based visualization of land use patterns, climate vulnerability, infrastructure development, and policy impacts.
11. Advanced analytics and decision-support tools for evaluating policy effectiveness and identifying emerging trends.
12. Policy simulation modules to assess the likely outcomes of proposed reforms before implementation.
13. Integration of satellite imagery, remote sensing, land records, socio-economic datasets, and geospatial databases.
14. AI-assisted research tools for trend analysis, literature synthesis, predictive modelling, and scenario analysis.
15. Innovation portal supporting hackathons, research grants, pilot projects, and knowledge competitions.
16. Interactive dashboards displaying: Research outputs, Policy performance indicators, Land use trends, Climate resilience metrics, Land dispute statistics, Project implementation outcomes, Geospatial insights
17. Secure role-based access for researchers, government officials, institutions, and public users with appropriate permissions.
18. APIs for seamless integration with existing government platforms, research databases, GIS systems, and digital governance initiatives.

The platform should establish a sustainable national ecosystem for applied research and policy innovation, enabling evidence-based decision-making, fostering interdisciplinary collaboration, accelerating technological adoption, and supporting resilient, transparent, and future-ready land governance in India.

Suggested component and Technology:

Component	Suggested Technology
Database	PostgreSQL, MongoDB, Elasticsearch
Data Analytics	Apache Spark, Pandas, NumPy
Document Management	Elasticsearch, Apache Solr, OCR Integration
Cloud Infrastructure	NIC Cloud (MeghRaj) / AWS / Azure Government Cloud

Collaboration Tools	Knowledge Portal, Discussion Forums, Workflow Management, Research Project Tracking
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Sub-theme: Digital Knowledge Management, Artificial Intelligence, Geospatial Technologies & Evidence-Based Policy Innovation for Land Governance.

Organization: Ministry of Rural Development, Department of Land Resources, PME Division.
